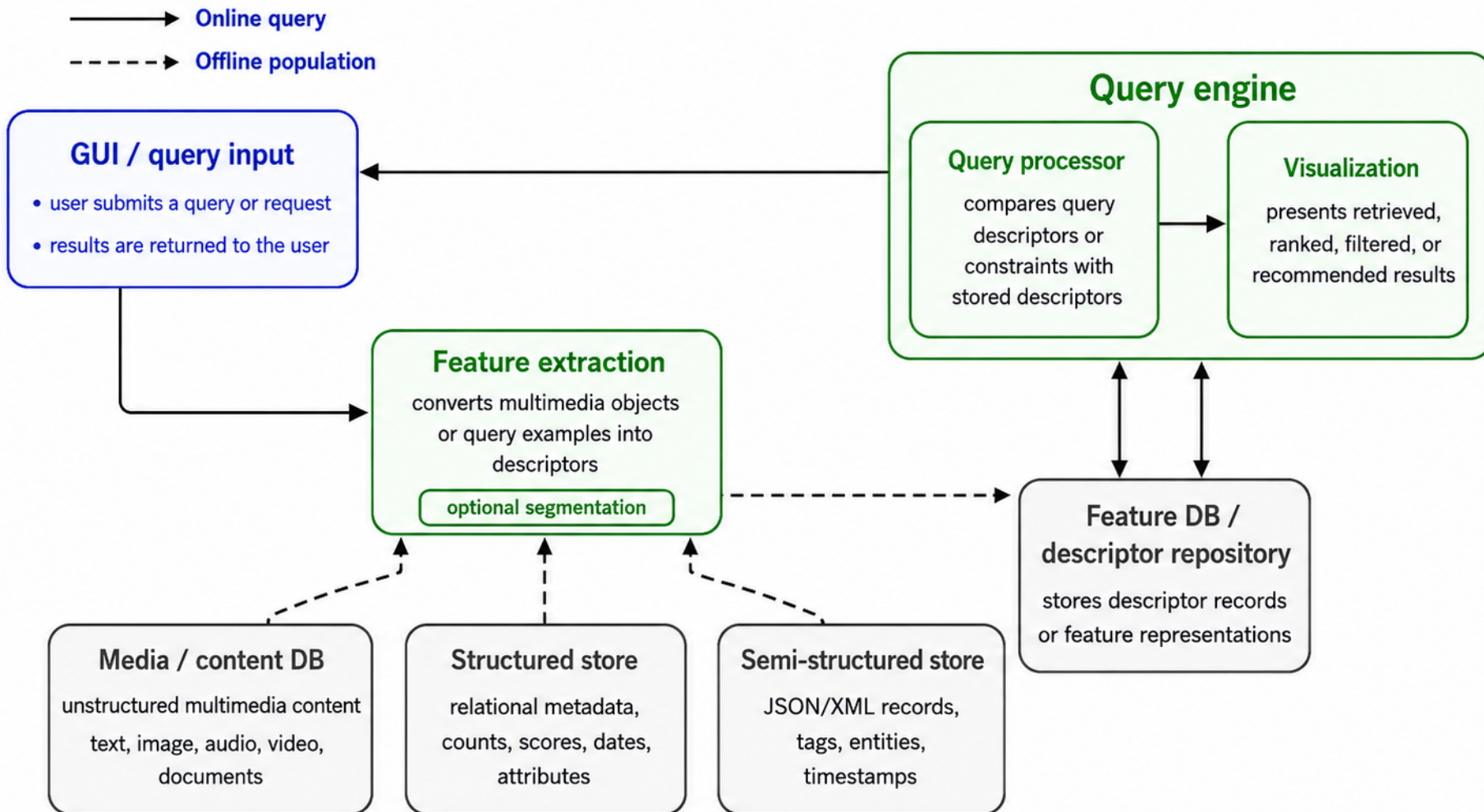


Reference architecture for multimedia data applications

Generic software modules — not tailored to one domain

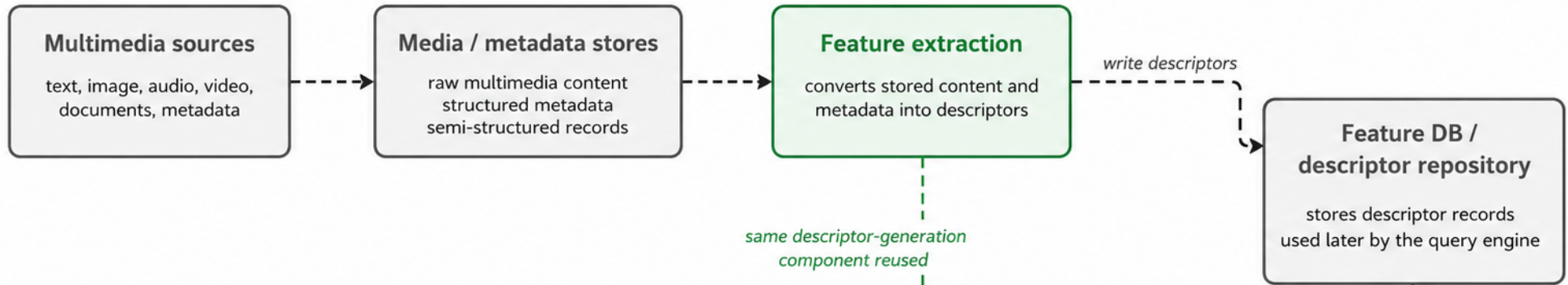


Software architecture: ingestion path and query path

How multimedia data and queries flow through the system

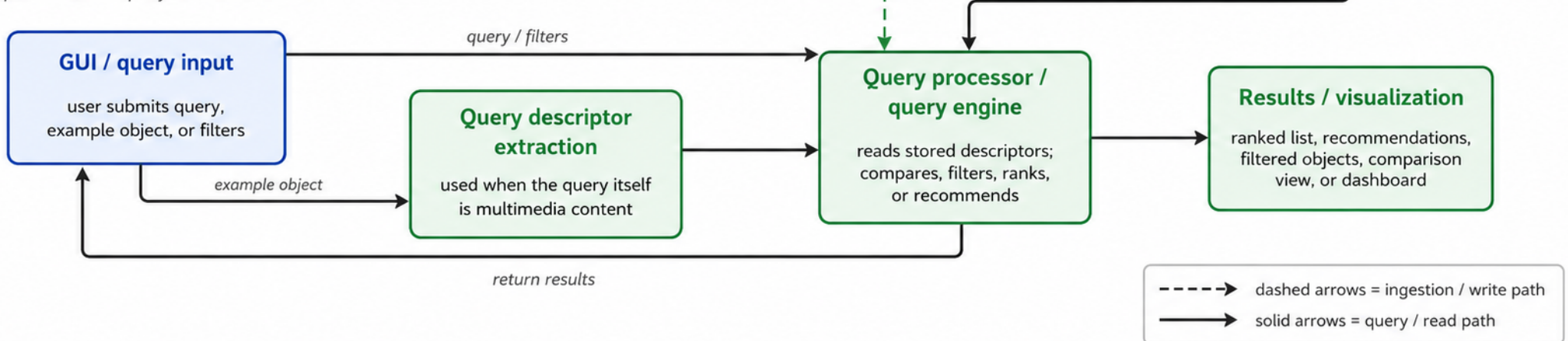
Offline ingestion path

populate the descriptor repository before user queries



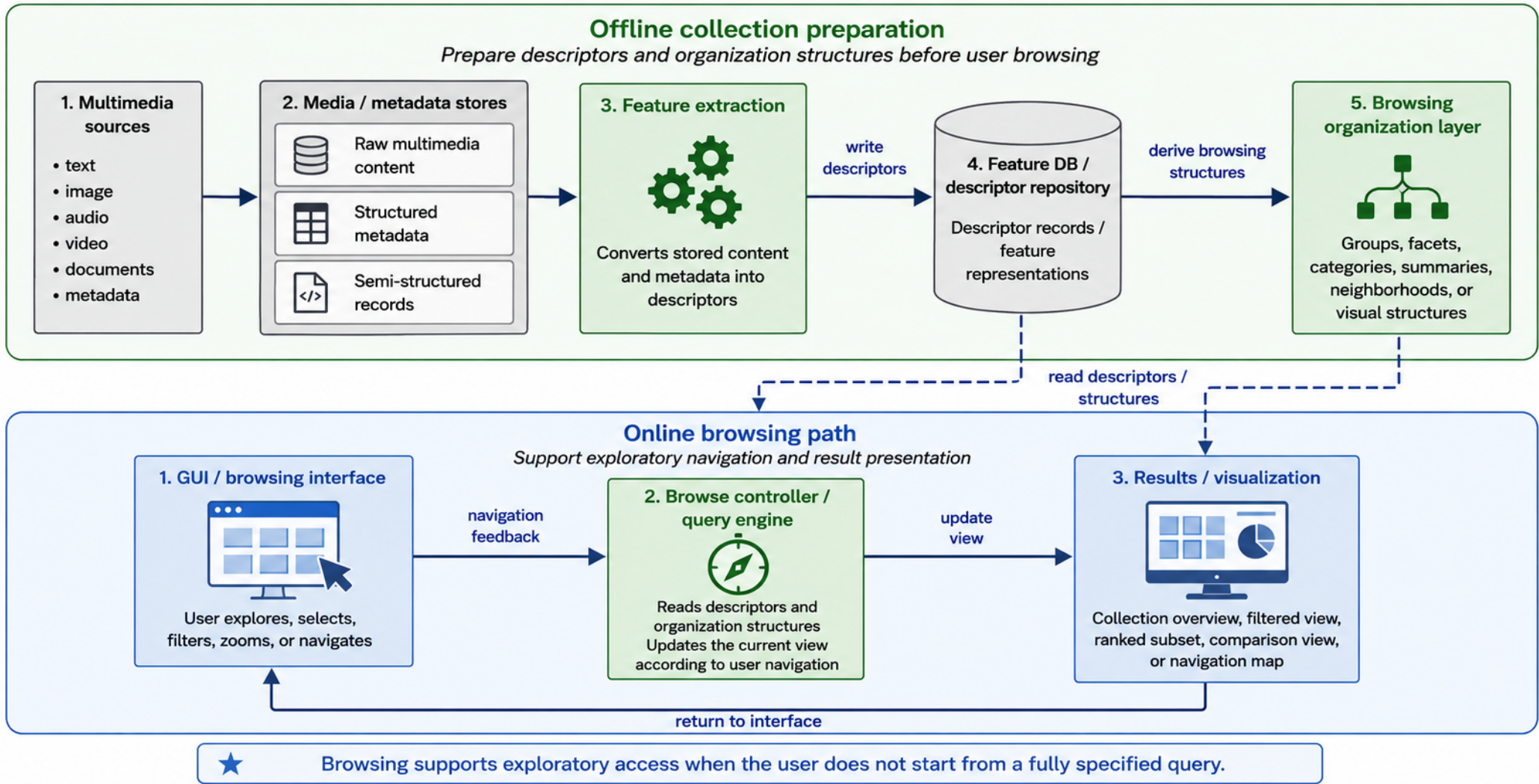
Online query path

process a user query and return results



Software architecture: browsing task

How users explore multimedia collections without a precise query

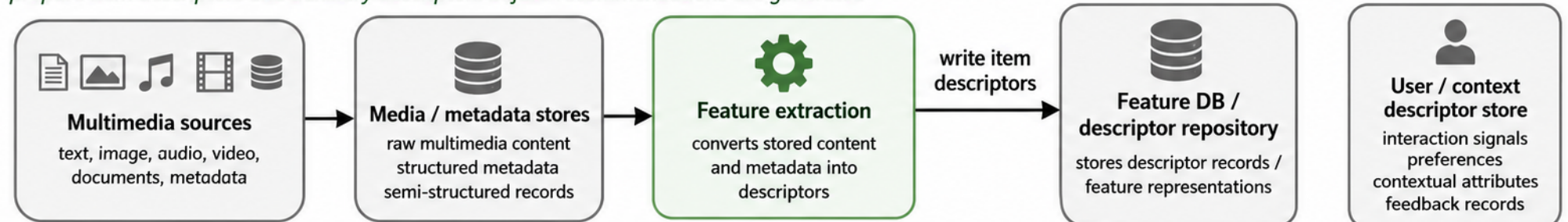


Software architecture: recommendation task

How multimedia descriptors support personalized or context-aware suggestions

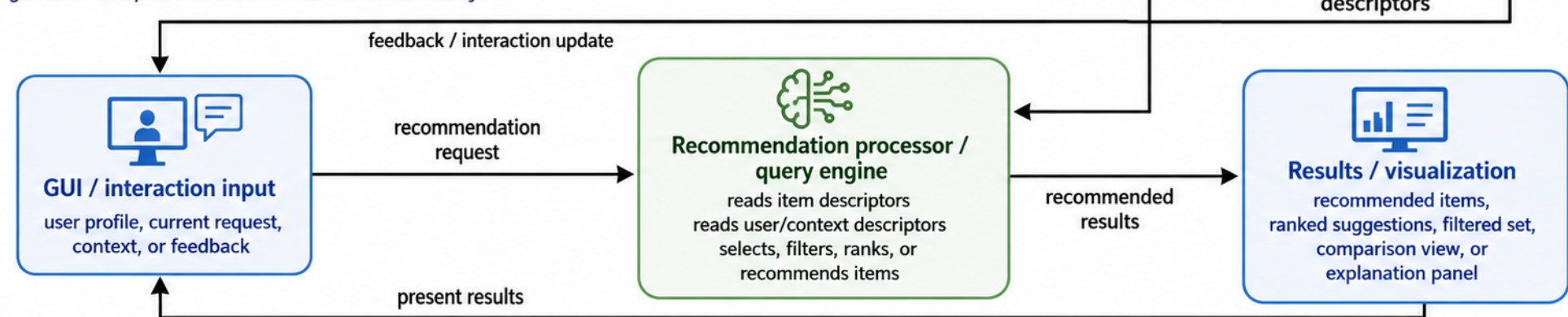
Offline descriptor preparation

prepare item descriptors and auxiliary descriptors before recommendations are generated



Online recommendation path

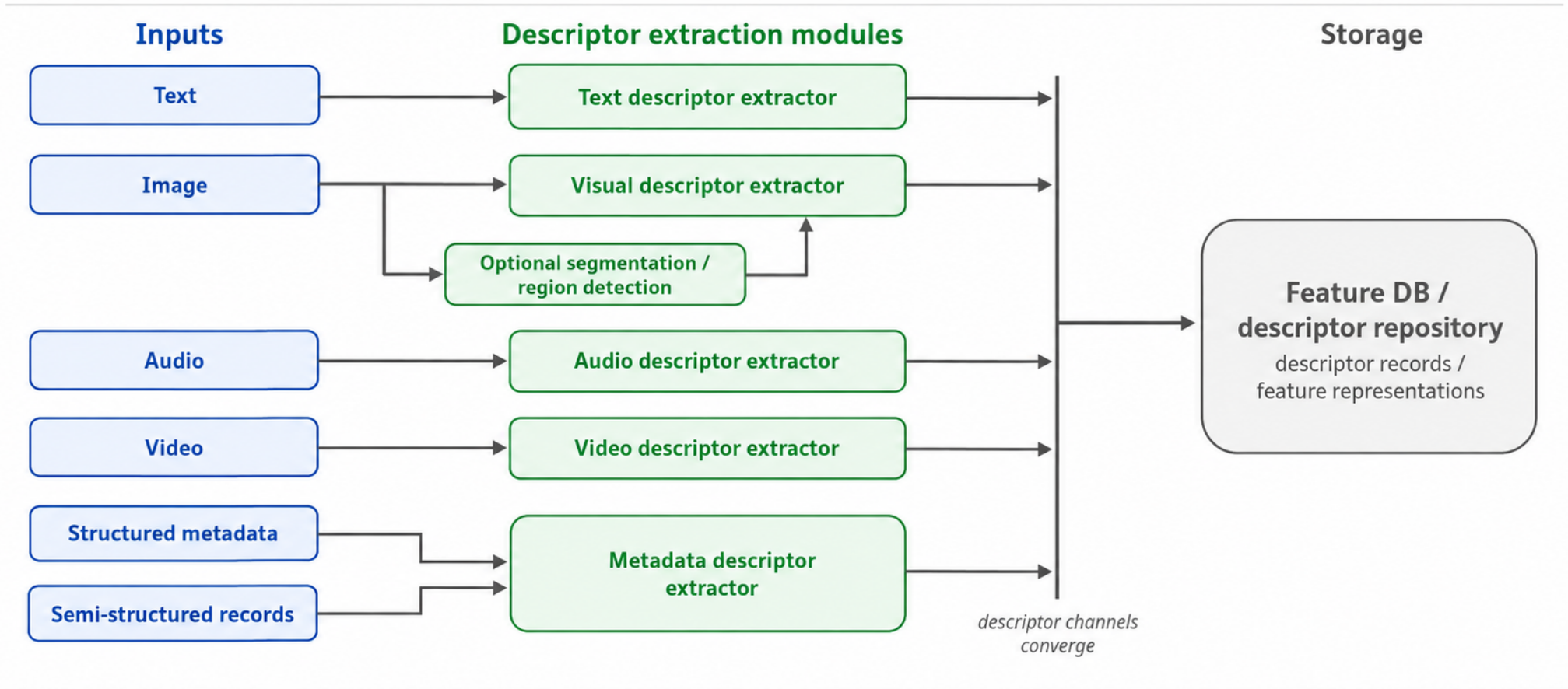
generate and present recommended multimedia objects



Recommendation reuses the same descriptor repository, but the online task starts from user/context signals rather than an explicit query.

Inside the feature extraction module

Generic descriptor channels for multimedia data



Each application activates only the channels it needs; the architectural template remains the same.